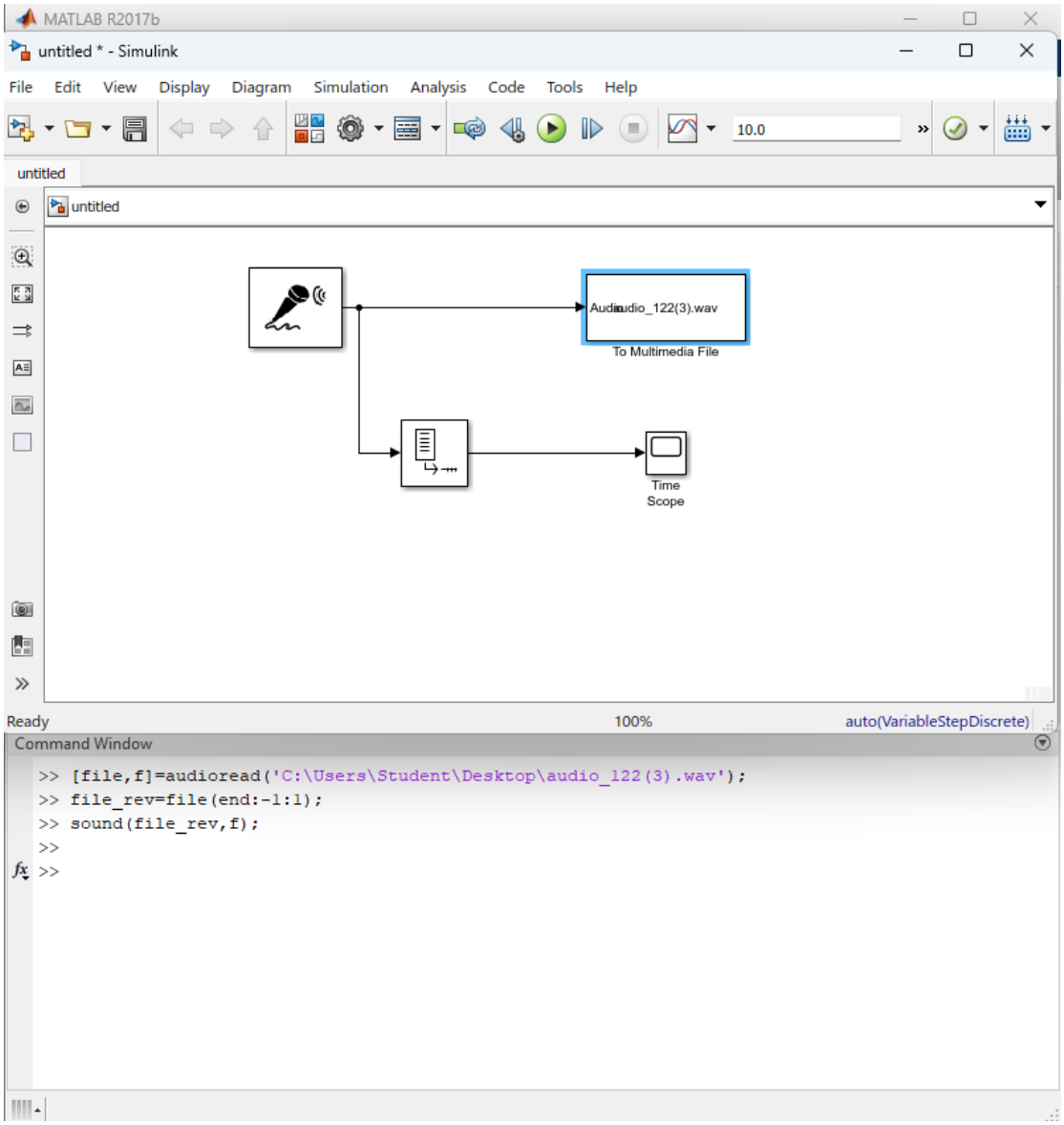
Roll No: CS23110

Lab 5

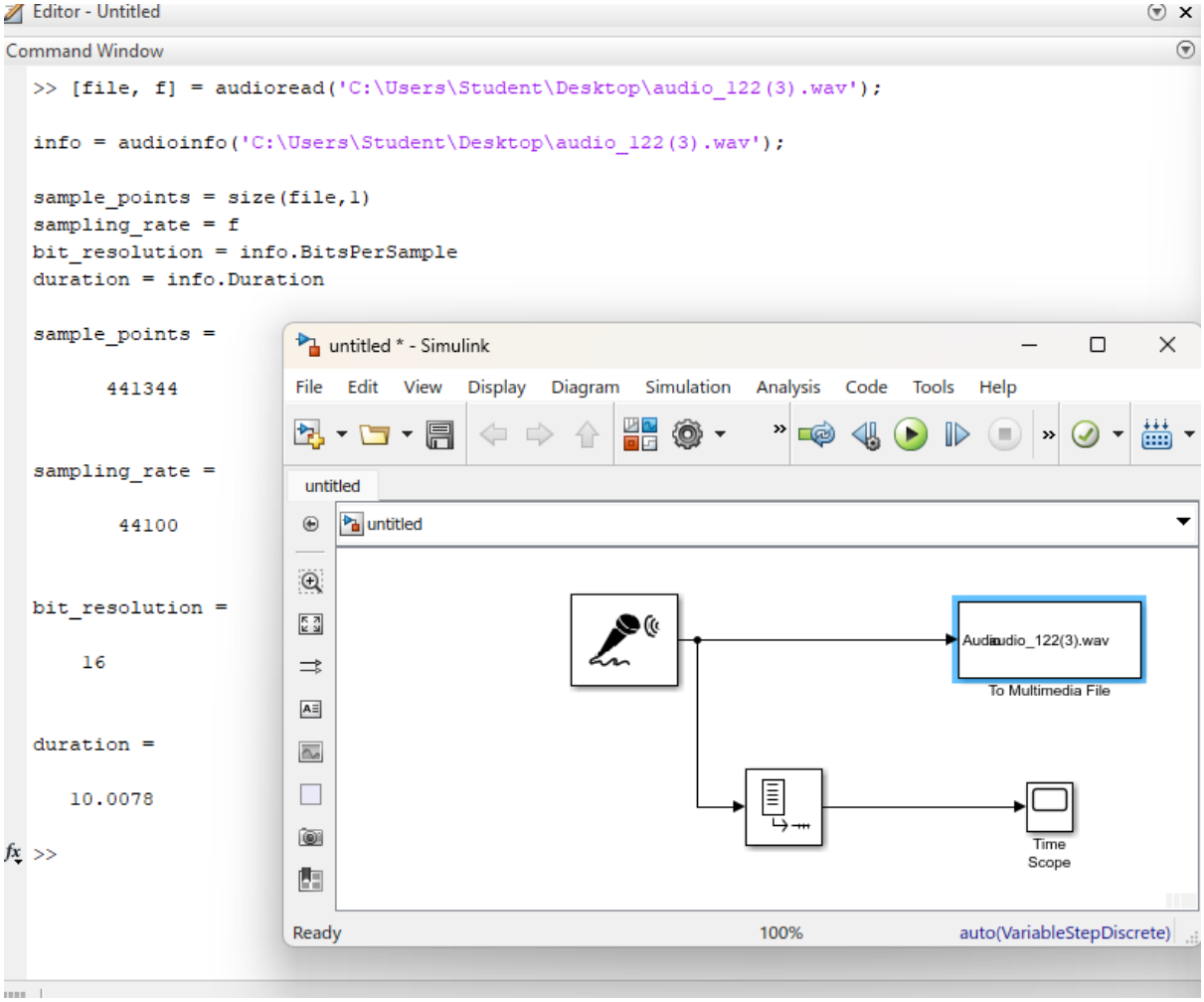
Task 1



Comments:

For the first task, I noticed that when I recorded "we" and reversed the audio array using the file(end:-1:1) command, it sounded surprisingly like the word "you". In the same way, playing the recording of "you" backward sounded very similar to "we". Because of this phonetic trick, reversing the full sentence "We are you" ended up sounding like "you are we".

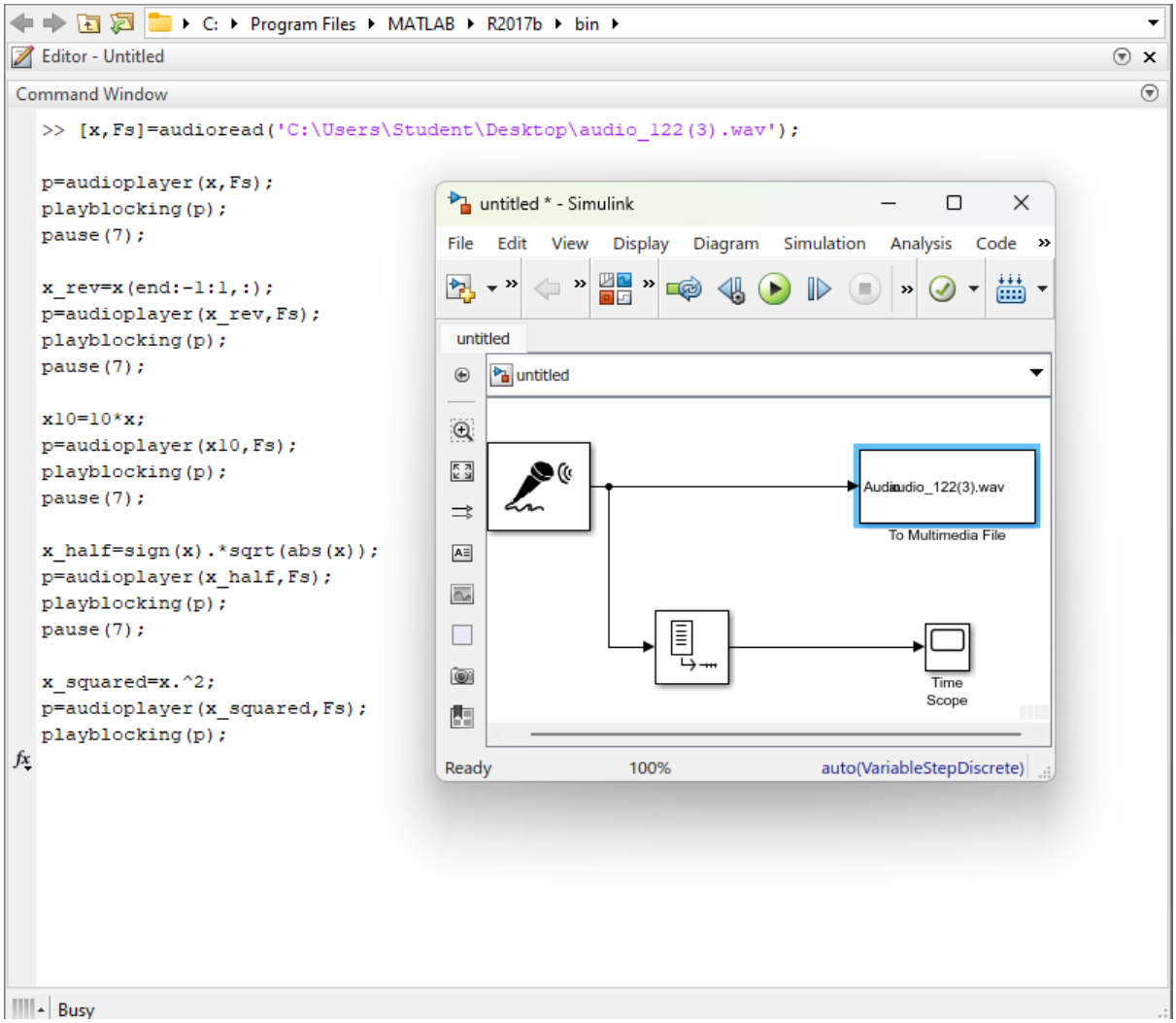
Task 2



Comments:

Using the audioinfo and size commands made it very easy to extract the exact properties of the audio file. My command window output showed that the file contained 441,344 sample points, had a sampling rate of 44,100 Hz, used a 16-bit resolution, and lasted for about 10 seconds.

Task 3



Comments:

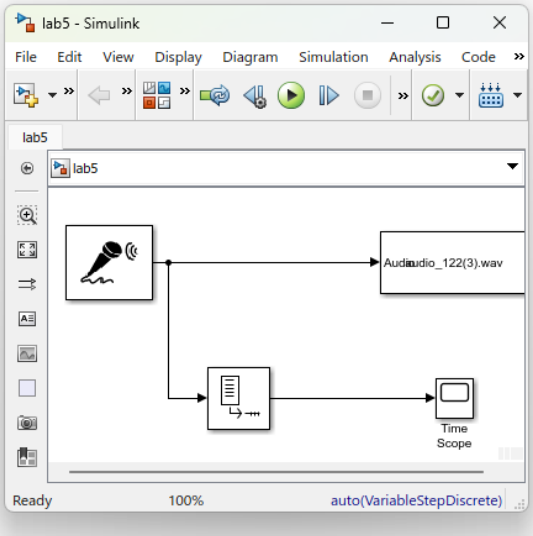
Task 3 Observations Applying different mathematical operations to the audio signal produced very distinct playback effects:

- **Multiply by -1:** The audio sounded exactly the same because flipping the signal upside down doesn't change the frequencies our ears pick up.
- **Reverse in time axis:** Using x(end:-1:1, :) played the sentence backward, making it sound like complete gibberish.
- **Multiply by 10:** Using 10*x simply increased the amplitude, making the playback much louder.
- **Replace with square root:** Using sign(x).*sqrt(abs(x)) added a lot of noise and made the recording sound static and distorted.
- **Replace with square:** Using x.^2 heavily distorted the voice, changing the pitch and making the words nearly impossible to recognize.

Task 4

```
>> [x,fs] = audioread('C:\Users\Student\Desktop\audio_122(3).wav');

disp('Original recording');
sound(x,fs);
pause(6);
% 32 kHz
x32 = resample(x,32000,fs);
disp('Playing 32 KHz');
sound(x32,32000);
pause(6);
% 16 kHz
x16 = resample(x32,16000,32000);
disp('Playing 16 KHz');
sound(x16,16000);
pause(6);
% 8 kHz
x8 = resample(x32,8000,32000);
disp('Playing 8 KHz');
sound(x8,8000);
pause(6);
% 4 kHz
x4 = resample(x32,4000,32000);
disp('Playing 4 KHz');
sound(x4,4000);
pause(6);
% 2 kHz
x2 = resample(x32,2000,32000);
disp('Playing 2 KHz');
sound(x2,2000);
pause(6);
% 1 kHz
x1 = resample(x32,1000,32000);
disp('Playing 1 KHz');
sound(x1,1000);
```



Comments:

By using the resample function to repeatedly halve the sampling rate, I observed a steady loss in audio quality. The intelligibility of the speech started to noticeably break down at 4 kHz. By the time the rate dropped to 2 kHz and 1 kHz, the audio was so muffled and degraded that the words became difficult to understand.

Task 5

Command Window

```
% Resample back to original rate for comparison
y_back = resample(y, Fs, Fs_new);

% Make same length
N = min(length(x), length(y_back));
x = x(1:N);
y_back = y_back(1:N);

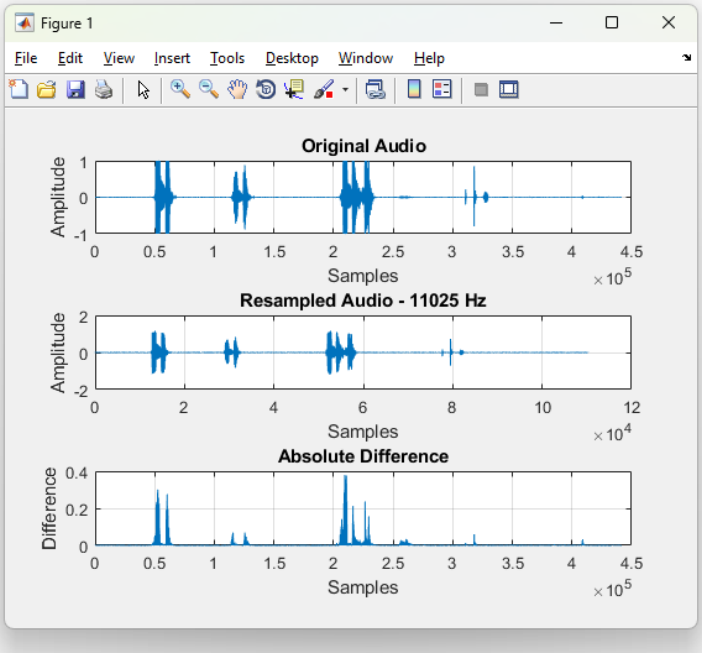
% Absolute difference
diff_signal = abs(x - y_back);

% Plot all 3 waveforms
figure;

subplot(3,1,1);
plot(x);
title('Original Audio');
xlabel('Samples');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(y);
title('Resampled Audio - 11025 Hz');
xlabel('Samples');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
plot(diff_signal);
title('Absolute Difference');
xlabel('Samples');
ylabel('Difference');
grid on;
```



Comments:

I used the subplot command to visually compare the original audio signal alongside the version resampled to 11025 Hz. When plotting the absolute difference between the two signals, the third graph showed clear spikes. These difference spikes visually prove that reducing the sample rate permanently alters and loses some of the original audio data.